to encode the compression and uncompression algorithms directly into integrated circuits. The potential of MPEG is vast. Think of thousands of video channels being carried on a single optical fiber running into your home. This is a key technology of the 21st century.

In addition to reducing the data rate, MPEG has several important features. The movie can be played *forward* or in *reverse*, and at either *normal* or *fast* speed. The encoded information is *random access*, that is, any individual frame in the sequence can be easily displayed as a still picture. This goes along with making the movie *editable*, meaning that short segments from the movie can be encoded only with reference to themselves, not the entire sequence. MPEG is designed to be robust to errors. The last thing you want is for a single bit error to cause a disruption of the movie.

The approach used by MPEG can be divided into two types of compression: *within-the-frame* and *between-frame*. Within-the-frame compression means that individual frames making up the video sequence are encoded as if they were ordinary still images. This compression is preformed using the JPEG standard, with just a few variations. In MPEG terminology, a frame that has been encoded in this way is called an intra-coded or **I-picture**.

Most of the pixels in a video sequence change very little from one frame to the next. Unless the camera is moving, most of the image is composed of a background that remains constant over dozens of frames. MPEG takes advantage of this with a sophisticated form of *delta encoding* to compress the redundant information *between frames*. After compressing one of the frames as an I-picture, MPEG encodes successive frames as predictive-coded or **P-pictures**. That is, only the pixels that have changed since the I-picture are included in the P-picture.

While these two compression schemes form the backbone of MPEG, the actual implementation is immensely more sophisticated than described here. For example, a P-picture can be referenced to an I-picture that has been *shifted*, accounting for motion of objects in the image sequence. There are also bidirectional predictive-coded or **B-pictures**. These are referenced to both a previous and a future I-picture. This handles regions in the image that gradually change over many of frames. The individual frames can also be stored out-of-order in the compressed data to facilitate the proper sequencing of the I, P, and B-pictures. The addition of color and sound makes this all the more complicated.

The main distortion associated with MPEG occurs when large sections of the image change quickly. In effect, a burst of information is needed to keep up with the rapidly changing scenes. If the data rate is fixed, the viewer notices "blocky" patterns when changing from one scene to the next. This can be minimized in networks that transmit multiple video channels simultaneously, such as cable television. The sudden burst of information needed to support a rapidly changing scene in one video channel, is averaged with the modest requirements of the relatively static scenes in the other channels.